INCH-POUND
MIL-P-15742C
2 September 1988
SUPERSEDING
MIL-P-15742B
20 February 1973
(See 6.5)

MILITARY SPECIFICATION

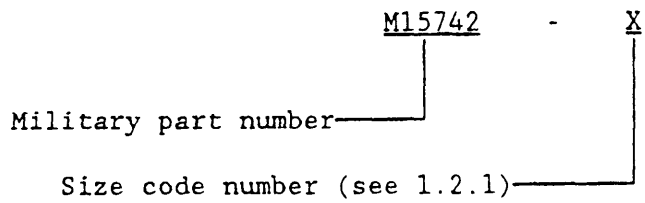
PLUGS, PLASTIC (HEAT-EXCHANGER-TUBE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heat-exchanger-tube plugs which are driven into both ends of leaking condenser or other heat-exchanger-tubes to prevent contamination of the fluid surrounding the tubes.

1.2 Classification. Plugs covered by this specification shall be designated in the following form (see 6.2 and 6.3):



1.2.1 Size code number. The size of the plug (see 3.3) is identified by a single-digit number (see table I).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4730
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the current version before use.

TABLE I. Size code number.

Plug size (inch)	Size code number
1/4	2
3/8	3
1/2	4
5/8	5
3/4	6

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY
MIL-P-116 - Preservation, Methods of.

STANDARDS

FEDERAL
FED-STD-406 - Plastics: Methods of Testing.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Plugs shall be made of laminated phenolic-material, consisting of layers of cotton fabric impregnated with natural-color phenolic resin and pressed under controlled conditions of temperature and pressure.

3.1.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extend practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Physical properties. The material from which the plugs are made shall conform to the following physical properties:

Tensile strength, flatwise pounds per square inch (lb/in^2), (min)	6,500
Compressive strength, flatwise lb/in^2, (min)	25,000
Impact strength, flatwise, Izod, foot pounds per inch of notch, (min)	2.3

3.3 Sizes. Plugs shall be of the following sizes, as specified (see 6.2), and shall conform to the dimensions shown on figure 1. Plugs shall not be out of round by more than 0.010 inch at any cross section along two-thirds of the length starting from the largest diameter end.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

Inch
1/4
3/8
1/2
5/8
3/4

3.4 Surface. Plugs shall be machined from laminated phenolic-material in such a manner that the long dimension of the plugs will lie parallel to the laminations. The edges shall be free from burrs or other defects. The surfaces shall be free from scratches, cracks, dents, or other imperfections which may prevent watertight seals between the plugs and the tube.

3.5 Water absorption. Plugs shall not show an increase in weight of more than 2.50 percent (see 4.4.2).

3.6 Loss on drying. The loss in weight of the plugs shall be not more than 4.0 percent of their original weight (see 4.4.3).

3.7 Expansion. The diameter of the plug at the large end shall increase not more than 4.40 percent (see 4.4.4). The change in diameter of the large end shall be determined both parallel and transverse to the laminations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection.

4.2.1 Lot. All plugs of the same size presented at one time shall be considered a lot for the purposes of sampling and inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

4.2.2 Sampling for visual and dimensional examination. Sample plugs shall be selected from each lot (see 4.2.1) in accordance with inspection level II of MIL-STD-105 and shall be subjected to the examination specified (see 4.3). The acceptable quality level (AQL) shall be 2.5 percent defective.

4.2.3 Sampling for tests. Sample plugs shall be selected from each lot (see 4.2.1) in accordance with inspection level S-2 of MIL-STD-105 and shall be subjected to the tests specified in 4.4.2 through 4.4.4. The AQL shall be 2.5 percent defective.

4.3 Visual and dimensional examination. The plugs shall be examined for conformance to the visual and dimensional characteristics specified in section 3 and figure 1.

4.4 Test methods.

4.4.1 Physical properties test. Prior to production, a material sample shall be tested in accordance with FED-STD-406 for conformance to 3.2.

4.4.2 Water absorption. The sample plugs shall be weighed and placed in a container containing approximately double the amount of 3 percent salt (sodium chloride) solution necessary to cover the plug. The solution shall be boiled continuously for 90 hours, the plugs remaining in the solution during the boiling period. The container shall be fitted with a reflux condenser to maintain a constant salt concentration during the boiling period. At the conclusion of the boiling period, the solution shall be allowed to cool for 4 hours and the plugs removed from the container. The water shall be wiped from the surface of the plugs and weighed immediately. The water absorption shall be determined as the difference of the final and the original weights.

4.4.3 Loss on drying. The sample plugs shall be weighed as received, and dried in an oven at 212 degrees Fahrenheit (°F) (100 degrees Celsius (°C)), for 16 hours. The plugs shall then be allowed to cool in a desiccator and weighed. The loss in weight shall be calculated from the difference of the original weight and the weight of the plugs after the drying period.

4.4.4 Expansion test. The diameter of the sample plugs shall be measured at their large ends, both parallel and perpendicular to the plane of the laminations. The plugs shall be boiled for 90 hours in the same container as that employed for the water absorption test (see 4.4.2). The diameters of the plugs shall again be measured at the points at which the initial measurements were taken.

4.5 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Plugs shall be cleaned and dried in accordance with MIL-P-116. Selection of the cleaning process and drying procedure shall be at the contractor's option.

5.1.1.2 Unit protection. Fifty plugs of one size, arranged standing on their ends alternately reversed, shall be unit packaged in a snug-fitting folding, paperboard box, conforming to PPP-B-566, variety 4. The applicable style, type and class shall be at the contractor's option. Unit protection shall meet the requirements of MIL-P-116, method III.

5.1.2 Level C. Unless otherwise specified (see 6.2), 50 plugs shall be unit packaged to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal packaging method may be used when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Plugs of one size (see 3.3), preserved as specified (see 5.1), shall be packed for shipment in snug-fitting fiberboard boxes conforming to PPP-B-636, class - weather resistant with variety, grade and style at the contractor's option. Method V, as specified in the appendix to the box specification, shall apply for box closure, waterproofing and reinforcing.

5.2.2 Level B. Plugs of one size (see 3.3), preserved as specified (see 5.1), shall be packed for shipment in snug-fitting fiberboard boxes conforming to PPP-B-636, class - domestic with variety, grade and style at the contractor's option.

5.2.3 Commercial. Plugs of one size (see 3.3), preserved as specified (see 5.1), shall be packed in containers acceptable to the common carrier and which will ensure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, and method of shipment shall conform to Uniform Freight or National Motor Freight Classification Rules or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2), interior packages and shipping containers shall be marked in accordance with MIL-STD-129. Item description shall include the plug size (see 3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The plugs are intended to be driven into both ends of leaking condenser or other heat-exchanger-tubes, thus preventing contamination of the surrounding fluid.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Military specification part number required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Size of plugs required (see 3.3).
(e) Level of preservation-packaging and packing required (see 5.1 and 5.2).
(f) When other than 50 plugs per unit package is required (see 5.1.2).
(g) Special marking (see 5.3).

6.3 Definitive Military part number. The Military part number is a definitive part number which corresponds to the size of plugs covered by this specification. The Military specification number and the size code number are combined to form the definitive Military part number (see 1.2).

6.4 Subject term (key word) listing.

Fabric, resin-impregnated cotton
Material, laminated phenolic

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

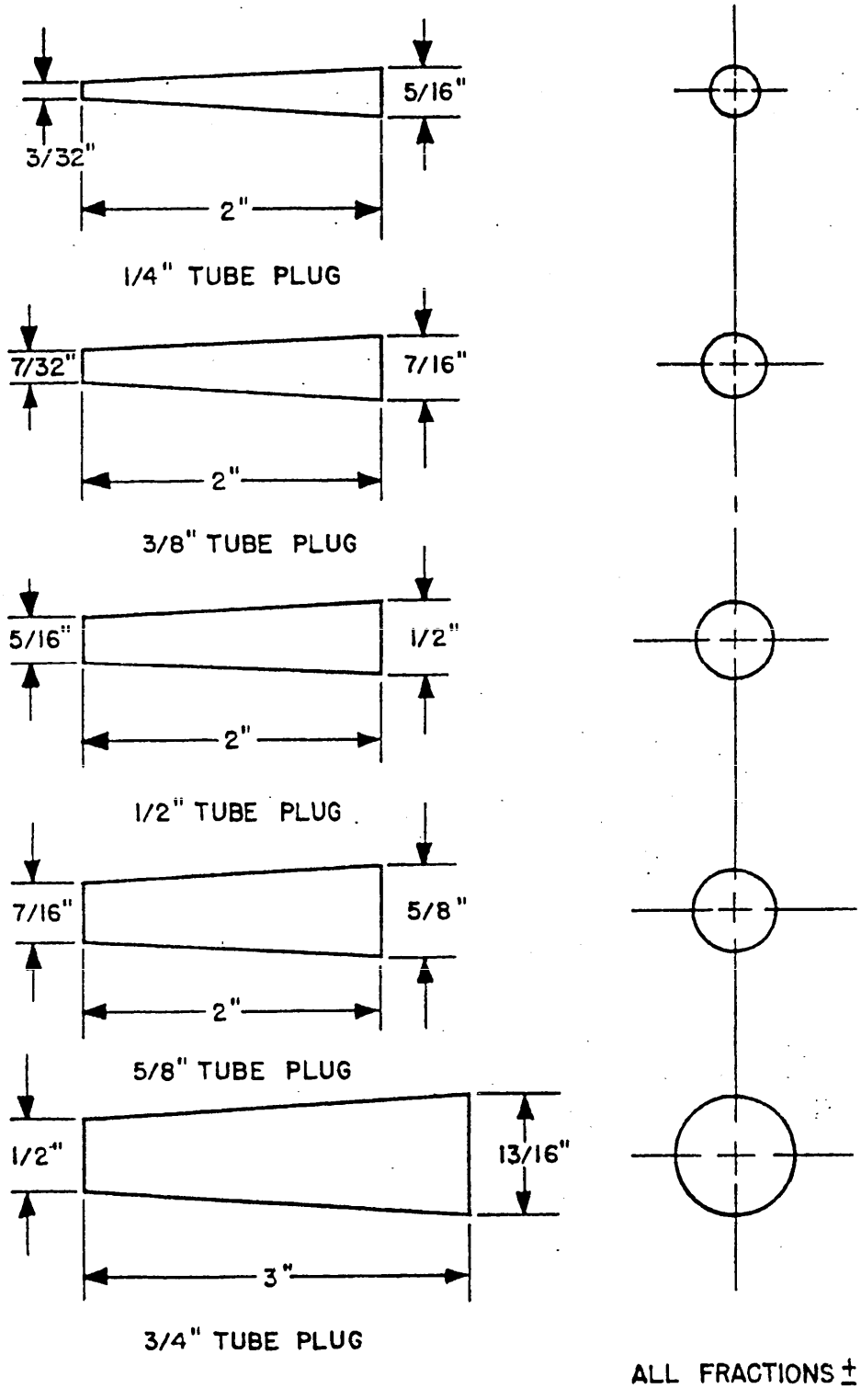
Custodians:
Army - ME
Navy - SH

Review activities:
Army - ME
Navy - SH
DSA - CS

User activities:
Army - WC
Navy - MC

Preparing activity:
Navy - SH
(Project 4730-1112)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.



SH 10489

FIGURE 1. Tube plug dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-15742C	PLUGS, PLASTIC (HEAT-EXCHANGER-TUBE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 2026-01-16T07:54Z
Check the source to verify that this is the current version before use.